

MUHAMMAD ZOHAIB

Aspiring Data Scientist | BS Data Science @ FAST NUCES Lahore

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PROFESSIONAL SUMMARY

Aspiring Data Scientist pursuing a BS in Data Science at FAST NUCES Lahore. Strong foundations in Python, C++, and machine learning with hands-on experience building intelligent, data-driven systems. Passionate about applying ML models and analytical pipelines to real-world business problems from fashion intelligence and business attribute prediction to enterprise decision-support platforms. Eager to contribute data science skills to drive measurable business outcomes.

EDUCATION

Bachelor of Science in Data Science

Expected July 2028

FAST NUCES, Lahore, Punjab

- CGPA: 3.35
- Core Courses: Data Structures & Algorithms, OOP (C++), Multivariate Calculus, Probability & Statistics, Python, Database Systems, Software Engineering

Intermediate (HSSC) — Computer Science

October 2024

Kips College, Lahore, Punjab

- Subjects: Computer Science, Physics, Mathematics | Grade: 89.9% — A+

Matriculation (SSC) — Computer Science

August 2022

Jauhar Foundation High School, Lahore, Punjab

- Subjects: Physics, Mathematics, Chemistry, Computer Science | Grade: 95.5% — A+

PROJECTS

CADS-Bridge — Collaborative Analytics & Decision Support Platform

2025

Web Application | FAST NUCES, Lahore

- Co-developed a web-based platform bridging Chartered Accountant (CA) and Data Science (DS) teams within enterprises, eliminating siloed workflows and enabling unified, data-backed financial decision-making.
- Designed role-based dashboards for CA teams (financial KPIs) and DS teams (model-driven metrics), with real-time cross-functional performance monitoring and side-by-side team comparisons.
- Implemented isolated firm-specific environments with role-based access control, ensuring zero data cross-contamination across organizations and full regulatory compliance.
- Built a Conflict Detection Engine that automatically flags discrepancies between DS revenue projections and CA actuals exceeding a 5% threshold, reducing decision-making delays.
- Developed a Collaborative Annotation Engine allowing CA and DS teams to annotate model outputs and financial statements inline with version-tracked comments and resolution threads.
- Engineered a Stakeholder Reporting Engine combining financial narratives and model insights into executive-level dashboards with automated generation, version control, and approval workflows.
- **Technologies:** Web Application, Role-Based Access Control, REST APIs, Database Design, Audit Trails, Real-Time Collaboration

Travel Buddy — Full-Stack Travel Platform

2025

Web Application | FAST NUCES, Lahore

- Designed and developed a comprehensive travel platform enabling users to plan and manage trips, find compatible travel buddies, and access a rich database of destinations, accommodations, and experiences.
- Implemented a Matching Algorithm suggesting compatible buddies based on budget, travel style, and interests, with a built-in expense-splitting module for accommodation, transport, and food.

- Architected an extensive hierarchical database (Countries → Cities → Attractions → Hotels → Restaurants) with user ratings, reviews, weighted trust scoring, and community-driven place suggestions.
- Integrated Google Maps/OpenStreetMap APIs for optimized route suggestions, multi-modal transport comparisons (flights, buses, car rentals), and offline map downloads for low-connectivity areas.
- Built vlogger-specific features including a content hub for video/photo uploads, marketplace integration for travel guides, and auto-generated documentation for visa applications and sponsorships.
- Incorporated an AI-Powered Itinerary Generator producing day-by-day trip plans based on user preferences, alongside real-time currency conversion and emergency assistance tools.
- **Technologies:** SQL, Database Design, Backend Development, API Integration, System Architecture

Fashion Image Classifier & Business Decision Intelligence

2024

Machine Learning & Computer Vision | FAST NUCES, Lahore

- Built a fashion intelligence system classifying clothing items from images and predicting business attributes: price range, sale likelihood, and selling rate bridging computer vision with commercial decision-making.
- Implemented a two-stage CV pipeline: YOLOv8 for clothing detection and localization, followed by HOG, GLCM, LBP, and Fourier Transform for extracting color, material, pattern, and size attributes.
- Developed and benchmarked two classifiers: Random Forest with handcrafted HOG and HSV features (87.56% accuracy, 10 min training) vs. a custom CNN with 3 convolutional blocks and batch normalization (92.25% accuracy).
- Engineered a feature mapping pipeline converting continuous CV outputs to categorical schema via color quantization, texture discretization, and pattern classification for downstream business prediction.
- Trained a Gradient Boosting Regressor achieving 78.3% price accuracy (within $\pm 15\%$), 81.7% sale likelihood classification, and 76.9% selling rate prediction from visual features alone.
- Deployed the complete pipeline as a Streamlit web application supporting image upload, live webcam capture, model selection, confidence scores, attribute visualization, and real-time business predictions.
- **Technologies:** Python, YOLOv8, CNN, Random Forest, Gradient Boosting, OpenCV, Streamlit, Scikit-learn, TensorFlow/Keras

Social Network Simulation — C++ Console Application

03/2025 – 06/2025

FAST NUCES, Lahore

- Developed a console-based social media simulator supporting user profiles, friend connections, and dynamic news feed generation for multiple user types.
- Applied core OOP principles, inheritance for differentiated user roles, encapsulation for data security, and polymorphism for dynamic content rendering across user types.
- Designed and implemented efficient data structures (arrays, vectors, linked lists) to manage user data, social connections, and feed generation at scale.
- **Technologies:** C++, OOP (Inheritance, Polymorphism, Encapsulation), Data Structures & Algorithms

Solar System Calculator — C++ Console Application

11/2024 – 02/2025

FAST NUCES, Lahore

- Engineered a C++ application calculating solar panel requirements and ROI for homeowners based on electricity consumption patterns and local weather data inputs.
- Implemented modular OOP architecture with dedicated classes for panel types, energy calculations, and user input, ensuring clean separation of concerns and easy extensibility.
- Utilized file I/O handling to persist and reload user profiles across sessions, enhancing overall user experience and enabling multi-session tracking.
- **Technologies:** C++, OOP (Classes, Objects), File I/O, Mathematical Modeling

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, HTML/CSS

Machine Learning & AI: Scikit-learn, TensorFlow/Keras, CNN, Random Forest, Gradient Boosting, YOLOv8

Computer Vision: OpenCV, HOG, GLCM, LBP, Object Detection, Feature Extraction

Data Science: Data Preprocessing, Feature Engineering, Model Evaluation, EDA

Core Competencies: OOP, Data Structures & Algorithms, Database Design, REST APIs, System Architecture

Tools & Frameworks: Streamlit, Jupyter Notebook, Git, GitHub, MS Excel, MS Word

LANGUAGES

English: Professional Working Proficiency

Urdu: Native / Bilingual